Surface-Mount Schottky Barrier Rectifiers

eSMP® Series



Top view Bottom view

SMF (DO-219AB)

Cathode ○—▶|—○ Anode

LINKS TO ADDITIONAL RESOURCES

3D Models

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	1.0 A
V_{RRM}	60 V
I_{FSM}	40 A
V_F at I_F = 1.0 A (T_A = 125 °C)	0.56 V
T_J max.	175 °C
Package	SMF (DO-219AB)
Circuit configuration	Single

FEATURES

AUTOMOTIVE GRADE Available

RoHS COMPLIANT HALOGEN FREE

- Low profile package
- Ideal for automated placement
- Low forward voltage drop, low power losses
- Low leakage current
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- Wave and reflow solderable
- AEC-Q101 qualified available
  - Automotive ordering code: base P/NHM3
- Compatible to SOD-123W package case outline
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

TYPICAL APPLICATIONS

For use in high frequency inverters, freewheeling, DC/DC converters, and polarity protection in commercial, industrial, and automotive applications.

MECHANICAL DATA

Case: SMF (DO-219AB)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-M3 - halogen-free, RoHS-compliant
Base P/NHM3 - halogen-free, RoHS-compliant, and AEC-Q101 qualified

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
M3 and HM3 suffix meets JESD 201 class 2 whisker test

Polarity: color band denotes the cathode end

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)			
PARAMETER	**SYMBOL**	**SS1FH6**	**UNIT**
Device marking code		16	
Maximum repetitive peak reverse voltage	V_{RRM}	60	V
Maximum average forward rectified current (fig.1)	$I_{F(AV)}$ (1)	1.0	A
Peak forward surge current 8.3 ms single half sine-wave $T_{J\,(init)}$ = 25 °C	I_{FSM}	40	A
Operating junction and storage temperature range	T_J, T_{STG}	-55 to +175	°C

Note

(1) Free air, mounted on recommended copper pad area

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Instantaneous forward voltage	I_F = 0.7 A	T_A = 25 °C	V_F (1)	0.60	-	V
	I_F = 1.0 A			0.64	0.70	
	I_F = 0.7 A	T_A = 125 °C		0.53	-	
	I_F = 1.0 A			0.56	0.61	
Reverse current	V_R = 60 V	T_A = 25 °C	I_R (2)	-	3	µA
		T_A = 125 °C		90	450	
Typical junction capacitance	4.0 V, 1 MHz		C_J	90	-	pF

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 5 ms

THERMAL CHARACTERISTICS (T_A = 25 °c unless otherwise noted)			
PARAMETER	**SYMBOL**	**SS1FH6**	**UNIT**
Typical thermal resistance	$R_{\theta JA}$ (1)(2)(3)	125	°C/W
	$R_{\theta JM}$ (1)(2)(3)	21	

Notes

(1) The heat generated must be less than the thermal conductivity from junction-to-ambient: $dP_D/dT_J<1/R_{\theta JA}$

(2) Device mounted on FR4 PCB, 2 oz. standard footprint

(3) Thermal resistance $R_{\theta JA}$ - junction to ambient; $R_{\theta JM}$ - junction to mount

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PREFERRED PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
SS1FH6-M3/H	0.015	H	3000	7" diameter plastic tape and reel
SS1FH6-M3/I	0.015	I	10 000	13" diameter plastic tape and reel
SS1FH6HM3/H (1)	0.015	H	3000	7" diameter plastic tape and reel
SS1FH6HM3/I (1)	0.015	I	10 000	13" diameter plastic tape and reel

Note

(1) AEC-Q101 qualified

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

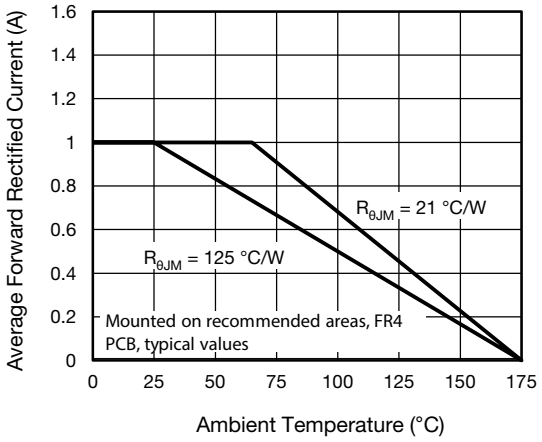


Fig. 1 - Maximum Forward Current Derating Curve

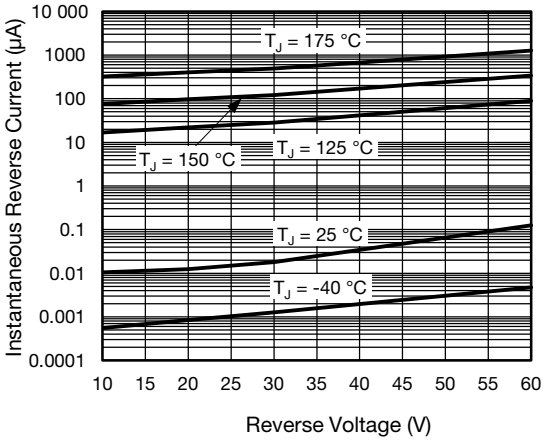


Fig. 4 - Typical Reverse Leakage Characteristics

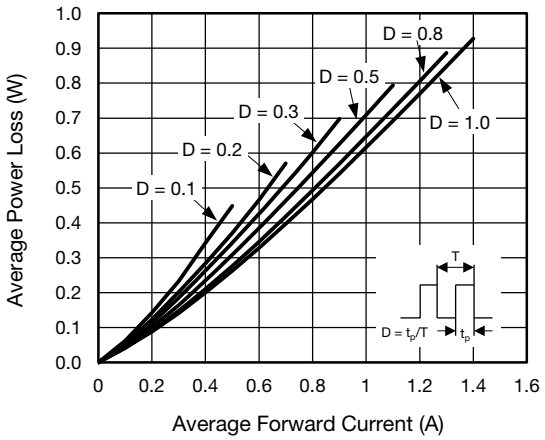


Fig. 2 - Average Power Loss Characteristics

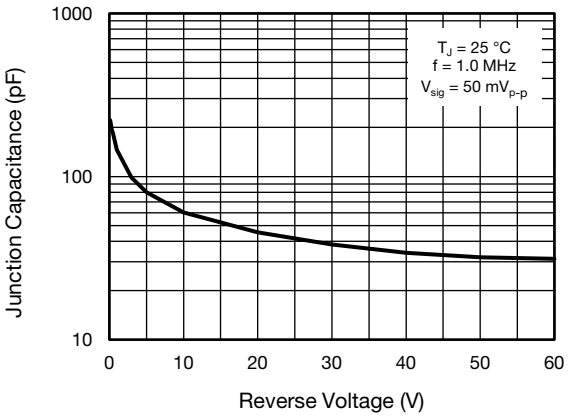


Fig. 5 - Typical Junction Capacitance

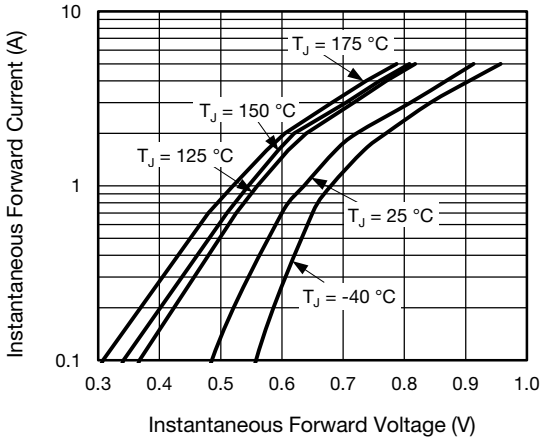


Fig. 3 - Typical Instantaneous Forward Characteristics

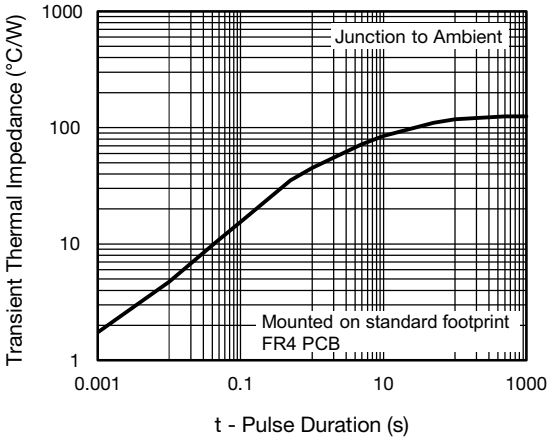
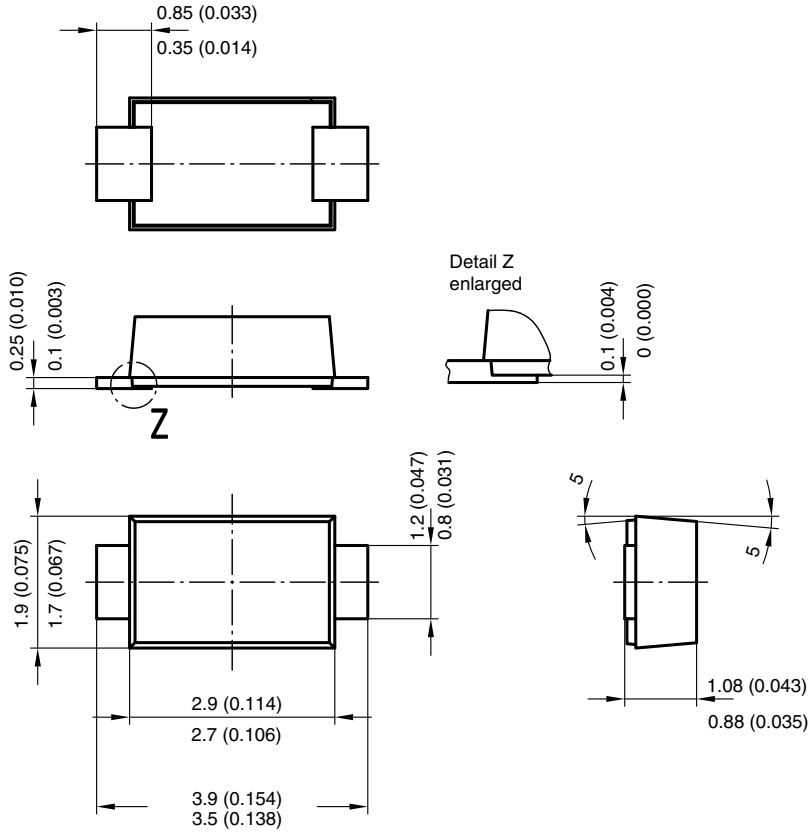


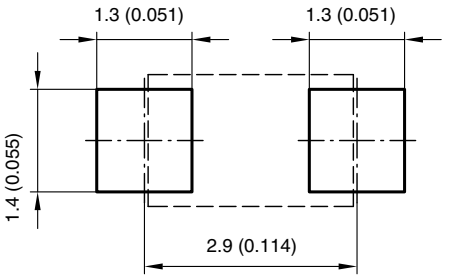
Fig. 6 - Typical Transient Thermal Impedance

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PACKAGE OUTLINE DIMENSIONS in millimeters (inches)



Foot print recommendation:



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